



Protecting the health of children with universal child cash benefits

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This Health Policy examines the relationship between child cash benefits and child health, with the goal of informing future policy development in the USA. As of 2024, more than 140 countries have adopted large-scale, government-funded child cash transfer programmes. High-income countries more often adopt universal or near universal programmes, while lower-income countries often impose means tests or condition benefits on specific behaviours. Evidence on the adoption of child cash benefits from a broad set of nations finds that they can improve a range of child health outcomes, with the most robust evidence of health benefits occurring when delivered to children younger than 5 years and during the prenatal period. During the 2021 expanded Child Tax Credit (CTC), the USA briefly joined other high-income countries by introducing a near universal, unconditional child cash benefit, which led to a historic decline in child poverty. Although the expanded CTC expired, state and local governments and communities have continued to advocate for and implement policies like it. On the basis of this success and building on global evidence, the USA should adopt a permanent child cash benefit consistent with other high-income countries and the 2021 expanded CTC. Nations further developing their cash benefits should also give special attention to the prenatal and infant period.

Introduction

There is much interest in the potential of cash transfers to improve the economic wellbeing and health of people. Cash transfers can take many forms, but generally are non-contributory (not accrued through taxed earnings), monetary transfers to households or individuals by government or non-government organisations.^{1,2} Increasingly, nations use cash transfers to advance a variety of social aims, with expansions during and after the COVID-19 pandemic. Many of these programmes were temporary or rolled out on a pilot basis.³

The evidence of effects on health is strongest for one kind of cash transfer: child benefits. Child cash benefit programmes typically make recurring payments of cash (as income or tax transfers) to households with children.⁴ As a form of social protection, government-run child cash benefits are an investment in the health, prosperity, and global competitiveness of a nation, with a

key objective to prevent and reduce child poverty. As of September, 2024, more than 140 countries (>70% of all countries) across every region of the world have adopted large-scale, government-funded child cash transfer programmes, many of which have been in operation for years or decades.⁵ Numerous studies draw conclusions about their effectiveness.

Approach

In this Health Policy, we examine the body of research on the relationship between child cash benefits and health on the basis of the global implementation of large-scale government programmes, with the goal of informing future policy development in the USA. Studies that fit these criteria often take advantage of variation in eligibility, comparing affected and unaffected populations before and after programme implementation, or changes in the timing of or value of benefits. The aim of this Health Policy is not to conduct a systematic review, but to use studies as evidence to inform future child cash benefit policy development in the USA and other countries.

While some reviews might give preference to randomised controlled trials (RCTs) because they offer the most certainty in detecting causal effects, there are concerns regarding external validity. Deaton and Cartwright highlighted that findings from RCTs are only directly applicable to the study sample, which is often only a convenience sample that differs from the broader population.⁶ Another challenge to RCTs is that the full effects of cash transfers might not be detectable in short-term studies, which many RCTs are, compared with other analytic designs, such as prospective cohort studies, which can follow a population over the lifecourse.

Another consideration is that when fielding cash transfer demonstrations in high-income countries, it would be difficult to assess the general equilibrium effects: the spillover, population-level effects of cash

Search strategy and selection criteria

We focused on studies of the relationship between child cash benefits and health, especially studies of the implementation of large-scale government programmes adopted by high-income countries. Studies were collected by the authors and research staff first by studying the references of widely cited studies and reviews in the field of child cash transfers and income support programmes. PubMed, Scopus, Overton, and Google Scholar were then searched from Jan 1, 2004, to March 4, 2024, using search terms, including “child cash benefit”, “child allowance”, “expanded child tax credit”, and “cash transfers” in conjunction with “child poverty” and “child health”. Only peer-reviewed studies, official government or government-affiliated reports, and working papers from high-impact institutions that house many well cited studies on relevant topics were considered. Only articles published in English were considered.

transfers. For example, a large-scale study in Kenya offered a cash transfer roughly equivalent to US\$1000 to approximately a third of all households in randomly selected rural villages. The study authors estimated that for every \$1 in cash benefits distributed, as that \$1 benefit circulated through the local economy, the effect on the host village was equivalent to \$2.50.⁷ Such a study in the USA or other high-income countries would be all but impossible because of the high cost of similar community-level saturation. The closest effort, to our knowledge, is Rx Kids (panel), a universal prenatal and infant cash transfer programme in Flint, MI, USA, which covers all expectant parents and infants in a small-sized city.

Reviewing studies that take advantage of the adoption of or changes to large-scale government child cash transfer programmes across many countries can inform policy discussions. Such studies, which use quasi-experimental methods or difference-in-differences analyses, must consider limitations to identifying causal effects. However, taken together, these studies provide the best evidence about the treatment of entire populations and offer the opportunity to analyse the mediating and moderating effects of community-level factors that are important for interpreting policy scalability. Such population-level evidence might be more applicable to the implementation of policy at scale than many experiments, which are often short in duration, small in scale, rarely universal, based on samples that are not representative of the population, and implemented by non-governmental organisations.

Child cash benefits and child poverty

Child cash benefits can be unconditional (with no terms) or conditional, for which families must adhere to specific

behaviours (such as school attendance or health-care use) to receive the transfer. Benefits can be means-tested and are only available to families whose income is less than a specific threshold or income level, or they can be universal or near-universal, meaning benefits are available to all or nearly all households with children. Near universal programmes exclude only the very wealthy, like in Denmark, or only apply to a restricted age range, like in South Korea, where benefits are paid to families with children aged 8 years or younger.⁴ Higher income countries often deploy universal or near-universal programmes, whereas lower-income countries tend to deploy conditional, contributory, or means-tested cash transfers.⁵

Globally, there are 49 countries with universal child cash benefits, nine with near-universal child benefits, and 87 with child benefits paid conditional to contributions or a means test.⁵ Table 1 reports on child cash programmes in Organisation for Economic Co-operation and Development (OECD) member countries; most programmes cover children from birth to age 18 years, although some programmes end at younger ages.⁴ The programmes offer a range of monthly benefits.

A goal of these programmes is to prevent and reduce child poverty, which has profound health and societal costs. Children living in poverty are more likely to have experiences of adversity, including chronic stress, obesity, food insecurity, poorer housing quality and housing instability, and elevated blood lead concentrations; inadequate access to health care, such as to a primary care physician or oral health provider; and inadequate access to human services, such as a therapist, all of which can lead to long-term health challenges, such as developmental delays, nutritional deficiencies, and

For more on Rx Kids see <https://rxkids.org>

Panel: Rx Kids, a universal and unconditional prenatal and infant cash prescription programme

To date, more than 1000 families are enrolled in Rx Kids, with more than US\$4 million prescribed. Building on global and domestic research, the Rx Kids evaluation team is examining whether the programme improves a wide array of individual and community outcomes, from maternal health to economic revitalisation. Child health outcomes of interest include maternal-infant health measures, such as improved birthweight and gestational age; greater health-care use; decreased child welfare involvement; improved maternal mental health and wellbeing; improved health maintenance, such as immunisations and screenings; and enhanced child enrichments. The evaluation design relies on both survey and administrative data, including through access to statewide vital records data. Such sources allow for a difference-in-differences design comparing trends in Flint, MI, USA before and after programme implementation to trends over the same period for similar cities in Michigan.

After just a few months of operation, a comparison of the number enrolled to the number of births documented at vital records indicates that the programme might be reaching nearly

100% of new families in Flint. Early evidence⁸ from preliminary difference-in-differences analyses comparing new mothers in Flint to those in the surrounding ineligible regions finds that the programme is associated with markedly increased housing security, improved ability to access food, and improved maternal mental health and wellbeing, including reduced postpartum depression.⁸

Led by a partnership between two public universities and administered by the international non-profit GiveDirectly, Rx Kids is a population-level programme that operates and informs at the scale of a large-scale government-run programme. Rx Kids is funded through a public-private partnership, including support from the State of Michigan using Temporary Assistance for Needy Family (TANF) block grant dollars. Michigan Governor, Gretchen Whitmer, championed⁹ the expansion of Rx Kids with an additional \$20 million in TANF block grant funds in the 2024–25 state budget dedicated to bringing the programme to other communities with high child poverty rates.

	Name	Benefit type	Age range, years*	Monthly amount† (US\$ equivalent)	Age with disability extension	Changes by age	Changes as family grows
Universal cash benefit programme							
Austria	Family allowance	Universal system with reduction based on income or assets	0–19	€114.00 (\$144.12)	No limit	Increases	Increases
Belgium	Basic family allowance	Universal system	0–19	€93.00 (\$124.25)	21 years	No	Increases
Estonia	Child allowance	Universal system	0–15	€60.00 (\$91.60)	No	No	Increases
Finland	Child allowance	Universal system	0–19	€94.88 (\$112.95)	No	No	Increases
France	Family allowance	Universal system with reduction based on income or assets	0–19	€65.98 (\$89.16)	No limit	Increases	Increases
Germany	Child benefit	Universal system	0–19	€204.00 (\$272.36)	No limit	No	Increases
Hungary	Family allowances	Universal system	0–19	HUF 12 200.00 (\$67.53)	No	No	Increases
Ireland	Child benefit	Universal system	0–15	€140.00 (\$190.48)	17 years	No	No‡
Israel	Family allowance	Universal system	0–19	ILS 152.00 (\$43.10)	No	No	Increases§
Italy	Family allowance	Social insurance	0–19	Variable	No	No	No
Latvia	Family allowance	Universal system	1–15	€11.38 (\$21.15)	19 years	No	Increases
Lithuania	Child benefit	Universal system	0–19	1.54 times higher than the monthly basic social benefit	No	No	No
Luxembourg	Family allowance	Universal system	0–19	€265.00 (\$312.50)	24 years	No	No
Netherlands	Child benefit	Universal system	0–17	€70.50 (\$88.46)	No	Increases	No
Norway	Child benefit	Universal system	0–19	NOK 1354.00 (\$111.98)	No	Decreases	No
Poland	Family allowances	Universal system with reduction based on income or assets	0–19	PLZ 95.00 (\$46.82)	No	Increases	No
Slovakia	Child allowance	Universal system	0–19	€24.95 (\$45.70)	No	No	No
Sweden	Child allowance	Universal system	0–15	SEK 1250.00 (\$141.77)	No	No	No
Near-universal cash benefit programme							
Aotearoa New Zealand	Family tax credit	Social assistance with reduction based on income or assets	0–19	NZ\$553.50 (\$374.75)	No	No	Decreases
Australia	Family tax benefit (part A and part B)	Social assistance with income test	0–15	AU\$191.24 (\$124.51)	No	Increases	No
Canada	Canada child benefit	Universal system with income test	0–19	Up to CA\$553.25 (\$453.48)	No	Decreases	No
Denmark	Child and youth benefit	Universal system with income test at high incomes	0–19	DKK 1532 (\$249.88)	No	Decreases	No
Iceland	Child benefit	Social assistance with reduction based on income or assets	0–19	ISK 19 542.00 (\$124.75)	No	No	Increases
Japan	Child allowance	Universal system with income test	0–15	¥15 000 (\$165.16)	No	Decreases	No
Portugal	Family allowances	Social assistance with asset and income test	0–23	€149.85 (\$262.43)	No	Decreases	Increases
Slovenia	Child benefit	Social assistance with income test	0–19	€20.36¶ (\$34.39)	No	No	Increases
South Korea	Childcare allowances	Universal system	0–8	KRW 484 000 (\$631.98)	No	Decreases	No
Switzerland	Child allowance	Social insurance	0–15	CHF 200.00 (varies by canton; \$199.60)	20 years	No	No
UK	Child benefit	Universal system with income test	0–15	£90.05 (\$132.04)	No	Increases	No

(Table 1 continues on next page)

Name	Benefit type	Age range, years*	Monthly amount† (US\$ equivalent)	Age with disability extension	Changes by age	Changes as family grows	
(Continued from previous page)							
Conditional, contributory, or means-tested cash benefit programme							
Chile	Family allowance	Non-contributory employment-related programme with income test	0–19	Insured earnings-related pay	No limit	No	No
Colombia	Family benefit	Social insurance with income test	0–19	Varies on the fund, minimum monthly payment is COP 23 386-00 (\$14-51)	No limit	No	No
Costa Rica	Family cash transfer	Social assistance with means test (unspecified); conditional on education	11–19	CRC 11 000-00 (varies by education programme; \$33-77)	No	No	No
Czechia	Child allowance	Social assistance with income test	0–19	CKR 500-00 (before income test; \$36-24)	No	Increases	No
Spain	Family allowance	Social assistance with means test	0–18	€28-42 (\$46-63)	No limit	No	No
Most recent available data at the time of writing collected by the International Labour Organization for the World Social Protection Report, 2024–26. ⁵ Monthly amount is the estimated amount paid monthly for a child aged 1 year. ¹⁰ Conversions to USD were made with the International Monetary Fund's implied purchasing power parity conversion factors for 2024. ¹¹ *Not including extensions for study or apprenticeships. †Given in the local currency unit, monthly amounts report the value of the benefit for the youngest child or first child if the benefit increases as families grow, but is not specifically age-related. ‡Ireland pays increments for twins and triplets. \$Amounts in Israel increase for the second, third, and fourth child, and returns to ILS 1526 from the fifth child onwards. ¶For one child living with both parents.							
Table 1: Child benefit programmes in Organisation for Economic Co-operation and Development member countries							

mental health disorders.^{12–15} Poor health in childhood is also associated with decreased school attendance and poor academic performance, which in turn restrict human capital.¹⁴ Healthier children are better equipped to succeed academically and professionally, in turn driving economic growth and reducing long-term public health expenditures.¹⁵ Thus, preventing child poverty—especially in early childhood—is a necessary precondition for a healthy and prosperous nation by proactively creating the socioeconomic conditions conducive to advancing physical and mental health for both children and their caregivers.

Universal and large-scale child benefits have been shown to be highly effective in reducing child poverty. A UNICEF analysis of comparable large-scale household survey data from the Luxembourg Income Study of 15 OECD member countries that have universal or near-universal child benefits finds that such programmes reduce poverty among households with children by 5 percentage points.^{4,16} For some countries, these benefits account for roughly half of the child poverty reductions from all cash transfers.^{4,16} Compared with other forms of cash transfers, universal child benefits attain high coverage rates, reaching a larger share of the child population with less administrative burden than programmes that impose means tests or behavioural conditions.^{4,16} In addition, universal programmes have broader public support and are more successful in reducing shame and stigma for those receiving benefits

than programmes that are more narrowly targeted.¹⁶ Due to their success in dramatically reducing child poverty, their high coverage rates, efficiency, and broad public support, UNICEF recommends universal, unconditional child cash benefits as the standard of care for child cash benefits globally. Indeed, UNICEF and the International Labour Organisation call child cash benefits “the foundational policy for child and social development”.⁴

Child cash benefits and child health globally

Flynn and colleagues offer a conceptual model for understanding how child cash benefits can improve child health.¹⁷ Child cash benefits have been shown to increase household income and decrease income instability, with particular effects on housing and food security, which have direct linkages to child health.¹⁷ Increased income and reduced income instability can reduce caregiver stress and cognitive load, which can in turn lead to fewer health-harming behaviours, such as smoking during pregnancy.¹⁸ Increased income and reduced cognitive load might also enable higher rates of health-care use and direct investments in children, such as more time spent engaging with them or enrichments to home learning environments. Investments made during early childhood have been shown to have long-term positive effects,^{19–22} because early childhood is a window for major brain development, when the foundation is laid for the long-term development of cognitive, social, and emotional skills.¹⁷

Studies on the effects of child benefits on the health of children and families in lower-income countries can inform US policy, even though the socioeconomic environments are different. These studies indicate that child cash benefits in lower-income countries are associated with improved access to health care, enhanced maternal and child nutrition, greater dietary diversity, and increased vaccination rates. Such benefits also appear to modify some intermediate determinants of health, such as pregnancy in adolescence and sexual behaviours that increase the risk of contracting sexually transmitted infections.^{1,23–25} Rasella and colleagues²⁵ found that higher coverage of the Bolsa Familia Programme, a large conditional cash transfer programme in Brazil, was associated with improved rates of mortality in children younger than 5 years (hereafter referred to as under-5 mortality), driven by declines in malnutrition-related deaths. Research finds that the Bolsa Familia Programme and the large-scale Family Health Programme both independently improved access to health care, vaccination rates, and maternal and child nutrition in Brazil, and had synergistic effects when combined.²⁵

Specific environmental conditions might enhance or attenuate the effects of child cash transfers.²⁶ Based on evidence from a preprint paper, a cash transfer programme in India was found to improve nutritional outcomes for all families who received cash and nutrition education, although improvements in height-for-age and weight-for-age were only present in communities with less open defecation.²⁷ Research on how contextual factors shape the efficacy of child cash transfers is growing and important.

Richterman and colleagues¹ merged mortality data from national household surveys in 37 low-income and middle-income countries (LMICs) to evaluate the effects of large-scale government cash transfer programmes on child mortality using a difference-in-differences approach. The authors found that cash transfer programmes were associated with significant declines in under-5 mortality (adjusted risk ratio [RR] 0·92, 95% CI 0·85–0·99).¹ Variation across countries and programme types indicate mortality improvements were greater when programmes offered a larger benefit, covered a larger share of the population, and took place in countries with lower starting health-care expenditures and lower life expectancy. The study found no difference in outcome based on whether programmes were conditional or unconditional, although larger coverage of the population was found to be associated with greater improvements in mortality rates, and unconditional programmes tend to have higher coverage than conditional programmes.¹ These findings indicate that such programmes work in a wide variety of environments, in settings with different cultures, and in different parts of the world.

Studies from high-income countries have also shown that child cash benefits can positively affect child health and development. Studies from Canada's child benefit

programmes show associations between expanded benefits and improvements in food security,²⁸ reductions in hunger, decreased aggressive behaviour among children, improved mental health, and improved educational outcomes.²⁹ Educational effects appear to be strongest for boys and mental health effects appear to be strongest for girls.²⁹ In Japan, families who receive child cash benefits have improved BMI, improved perceptions of health, and reduced behavioural problems compared with families who do not receive them.³⁰ In some studies, effects are restricted to specific subgroups, such as children from low-income households or impacts by sex, such as just for female children.^{31,32}

Research on child cash benefits in high-income countries shows that they are associated with increased spending on children's enrichment and educational activities, particularly for low-income populations, although magnitudes vary depending on the study.^{33,34} Evidence also suggests other changes to parental spending that improve the conditions in which children are raised, such as swapping restaurant spending for groceries,³⁵ increased ownership of durable goods (eg, telephones or vehicles)³⁶ and increased spending on non-durables (eg, childcare).³⁵ Researchers consistently find no evidence of increased spending—or even a decrease in spending—on tobacco and alcohol.^{34,35}

Prenatal cash benefits

There is a growing body of evidence supporting the fetal origins hypothesis, which highlights the importance of the prenatal period in shaping the lifecourse trajectory.³⁷ Poor and inadequate nutrition, stress, and exposure to toxins can adversely affect fetal development during this time of rapid cell division and differentiation. Hunter and colleagues³⁸ concluded that optimal nutrition not only supports fetal growth, but might also reduce the likelihood of preterm birth through strengthening the immune system and reducing oxidative stress. In addition to undernourishment, infection, illness, and exposure to toxins can lead to low birthweight or prematurity.³⁸ High cortisol caused by stress during pregnancy can also lead to prematurity.³⁸ Cash transfers during the prenatal period therefore have the potential to positively affect these mechanisms by improving nutrition and housing security, reducing stress, and improving prenatal health-care use.

In LMICs, evidence on the effects of cash transfers during pregnancy is growing. In Brazil, researchers found that the Bolsa Familia Programme was associated with a 17·8 g increase in birthweight, and that the risk of low birthweight was 11% lower for programme participants than non-participants.³⁹ A study of a conditional cash transfer programme called Oportunidades (also referred to as Progresora or Prospera) in Mexico found that participation in the programme (combined with improvements in the quality of prenatal care) was associated with a 127·3 g higher average birthweight compared with non-participants.⁴⁰

Table 2 reports on studies of cash transfer programmes including a prenatal payment in high-income countries, finding that they are associated with improved birth outcomes, such as greater birthweight, higher Apgar scores, and improved parental reports of wellbeing.^{42,49,53,58,59} In England and Wales, the roll-out of a universal child benefit in pregnancy was associated with increased birthweight of approximately 8–12 g, a reduction in the risk of low birthweight of 3–6%, and decreased risk of prematurity of 9–11%.⁵⁷ Mothers younger than 25 years living in economically deprived communities saw the greatest improvements.⁵⁷ A different study, however, of the same programme, which used a different design that could not account for the temporal effects of a concurrent global recession and the 2009 H1N1 outbreak, did not find that the programme affected birthweight.⁵⁶ The roll-out of a broad cash transfer programme in South Korea during the COVID-19 pandemic was associated with a 7.4 g increase in birthweight among children born to parents who received the benefit for an older child, after accounting for pandemic-related birthweight losses.⁵⁵ This increase was roughly in line with the weight gains associated with the programme in England and Wales.⁵⁷

Enns and colleagues⁵³ studied an unconditional prenatal income supplement programme for First Nations women with low incomes in Manitoba, Canada, comparing thousands of individuals who received the benefits between 2003 and 2011 with a matched comparison cohort who did not. The authors found that those receiving the prenatal cash benefit had reduced risks of low birthweight and prematurity. Mothers who received the benefit were more likely to start breastfeeding and vaccinate their children at ages 1 year and 2 years than those who did not receive the benefit. In addition, children in the programme were less likely to have language and cognitive deficits and had improved measures of kindergarten or preschool readiness compared with matched children not in the programme.⁵³ An earlier study of the same programme on 10736 women who received the transfers found similar effects.^{53,52} In addition, Enns and colleagues found a 10% reduction in the risk of having a child born small for gestational age.⁵² Finally, a study of a €2500 and universal prenatal cash transfer programme in Spain using a regression discontinuity design indicated that the cash transfer led to an 83% reduced incidence of very low birthweight among families with lower incomes.⁵⁴ Taken as a whole, there is a growing evidence base to support the use of cash transfers during pregnancy, from both low-income and high-income countries.

Income transfers and child health in the USA

Research on the implementation of or changes to cash and near-cash transfer programmes in the USA offers domestic evidence that such programmes improve child health. A study by Bailey and colleagues⁶⁰ followed

population-wide exposure to the county-by-county roll-out of the Food Stamp Program (now called the Supplemental Nutrition Assistance Program) in the USA between 1961 and 1975. Although restricted to the purchase of food, researchers find that such benefits are largely treated equivalently to cash by beneficiaries with other cash incomes.⁶¹ The authors linked 17.5 million Americans to their records in the 2000 Decennial Census and American Community Survey data, a nationally representative US Census Bureau household survey.⁶⁰ Exposure to the programme before age 5 years was linked to a 1.2 year increase in life expectancy, increased human capital (in years of schooling, educational attainment, and professional occupation), and improved economic self-sufficiency in adulthood.^{60,62}

Another long-term longitudinal study in the USA compared families with male children who received the USA's first government cash transfer programme for widows with children—the Mothers' Pension Program—while it was in effect between 1911 and 1935. The study was on families who applied and were initially deemed eligible, but who were later denied aid, which yielded a sample size of more than 16 000 male children living in 11 US states.⁶³ The authors found that male children of families approved for the benefits lived almost 1 year longer (the children from the lowest income families lived 1.5 years longer), were less likely to be underweight, attained a third of a year more schooling, and saw increased income in early adulthood compared with male children whose families were not approved for the benefits.⁶³

A series of studies exploited variation over time and across states in refundable tax credits—the Earned Income Tax Credit (EITC) and the Child Tax Credit (CTC)—to examine outcomes related to child health. Both the EITC and CTC are essentially lump sum cash transfers targeting families with children, conditional on earnings. Hoynes and colleagues⁴⁹ leveraged changes in the EITC across states and time and found that a \$1000 increase in the EITC benefit per family per year led to a 1.6–2.9% decline in low birthweight. Hamad and Rehkopf⁴⁸ used an instrumental variable approach and found small but significant improvements in birthweight among those who became eligible for increased values of the EITC. Strully and colleagues⁴⁶ found improvements in birthweight and reductions in rates of prenatal smoking (a partial explanation for improvements in birthweight) among unmarried women who only completed secondary education. Three additional studies also found improvements in birthweight associated with state EITC expansions.^{47,50,51} Braga and colleagues⁶⁴ found that an additional \$100 in average EITC benefit over childhood was associated with a 2.6% improvement in perceived child health and a 4.1% decrease in the likelihood of being obese in early adulthood (ages 22–27 years). Finally, large-scale RCTs during the 1960s explored the effects of a change to a Negative Income Tax cash transfer in the USA from the

	Country	Study design	Findings
Kehrer et al (1979) ⁴¹	USA	Randomised controlled trial of Gary Negative Income Tax	Babies in recipient families had a 117·6–540·4 g gain in birthweight
Chung et al (2016) ⁴²	USA	DID analysis of APFD	US\$1000 (around \$3270 in 2024) was associated with an increase of 17·7g in birthweight for babies of APFD recipients and a decrease of 8–14% in the rate of low birthweight births
Wyndham-Doud et al (2024) ⁴³	USA	Regression analyses on the APFD	Models adjusted to include birth selection and demographics show a small negative effect of a large exogenous cash transfer during pregnancy on birthweight
Ruffini (2023) ^{18*}	USA	DID and triple-difference estimates of economic impact payments and expanded child tax credit	An additional \$1000 in the benefit is associated with a 1·6–3·5% decrease in the prevalence of low birthweight births; payments during the third trimester had a much greater effect than earlier prenatal payments
Lyu et al (2024) ^{44*}	USA	Regression analysis of economic impact payments and expanded child tax credit	The effects of \$1000 during pregnancy on birthweight were not statistically nor clinically significant
Cho (2011) ⁴⁵	USA	DID on effects of PROWRA for immigrant families	PROWRA made some undocumented immigrants ineligible to receive most federal aid and was associated with an increase of 3·1 infant deaths per 1000 livebirths; the rate increased to 5·8 in financially disadvantaged groups
Strully et al (2010) ⁴⁶	USA	DID by state, fixed-effects model on EITC payment size	Presence of a state EITC is associated with a 15·7 g increase in birthweight, or 12·5 g after adjusting for smoking; effects were largest for women aged 19–34 years with less socioeconomic risk
Komro et al (2019) ⁴⁷	USA	Quasi-experimental DID analysis of state-level EITC laws	For Black mothers, presence of a state EITC was associated with a 16·12–37·12 g increase in birthweight (0·5–1·2%) compared with a 9·38–28·4 g increase (0·3–0·9%) for White mothers; state EITC is associated with a decreased likelihood of low birthweight with a decrease of 5–11·7% for Black families and 3·3–11·7% for White families, depending on benefit size
Hamad et al (2015) ⁴⁸	USA	Multivariable regression and instrumental variable analysis on EITC payment size	\$1000 increase in EITC was associated with a statistically insignificant increase in birthweight of 65·1g (p<0·1; 95% CI –0·46 to 130·6)
Hoynes et al (2015) ⁴⁹	USA	Quasi-experimental DID analysis of the OBRA93 expansion	\$1000 treatment-on-the-treated led to a 1·6–2·9% decline in the rate of low birthweight for unwed, less-educated mothers; the \$1000 increased the average birthweight and effects were larger for Black recipients (18·7 g increase) than White recipients (2·8 g increase); improvements in prenatal care, gestational length, 5-min Apgar scores, weight for gestational age, and reductions in drinking and smoking
Wicks-Lim et al (2017) ⁵⁰	USA	Quasi-experimental DID analysis of New York State and City's EITC, with a focus on ecological effects	A 15 percentage-point increase in EITC rates is associated with a 0·45 percentage-point decrease in neighbourhood-level prevalence of low birthweight for low-income neighbourhoods in New York City
Markowitz (2024) ⁵¹	USA	Quasi-experimental DID analysis of state-level EITCs and their generosity	Among low-educated women, states with an EITC are associated with an increase in birthweight by 9–27 g; increased generosity of state EITCs is associated with reduced prevalence of low birthweight (4–11% reduction)
Brownell et al (2016) ⁵²	Canada	DID analysis of HBPB	Participants (low-income maternal-infant dyads) had a 29% lower risk of low birthweight births (95% CI 0·63 to 0·81), a 24% lower risk of preterm birth (0·81 to 0·99) and a 10% lower risk of having a small for gestational age birth (0·81 to 0·99); reductions in prevalence of low birthweight (21%) and preterm births (17·5%); 13% increased risk of large for gestational age birth (1·05 to 1·23)
Enns et al (2021) ⁵³	Canada	Propensity matched cohort study of HBPB	First Nations women who received HBPB had a 26% reduced risk of low birthweight (95% CI 0·62 to 0·88) and 23% reduced risk of preterm births (0·68 to 0·88); an increased likelihood of initiating breastfeeding (adjusted risk ratio 1·05; 95% CI 1·01 to 1·09) and vaccinating children at age 1 year (1·10; 1·06 to 1·14) and 2 years (1·19; 1·13 to 1·25); children were also less likely to be considered developmentally vulnerable when they reached kindergarten
González et al (2022) ⁵⁴	Spain	Regression discontinuity design of universal child benefit	Among recipients of the transfer, they found a significant decrease in the rate of very low birthweight births (decline of 62% for the full sample and 83% in the lower income sample); effects were greatest for disadvantaged, preterm babies: 101% decrease in very low birthweight, 27% decrease in low birthweight, and the average birthweight increased by 228 g
Jung (2023) ⁵⁵	South Korea	DID analysis of natural experiment, COVID-19 relief	Babies born during the COVID-19 pandemic had an average birthweight 9·6 g lower than pre-pandemic cohorts; the universal or unconditional cash transfer increased the mean birthweight by 7·4 g, making up 80% of the pandemic-related decrease
Leyland et al (2017) ⁵⁶	Scotland	Interrupted time-series analysis of the natural experiment of the Health in Pregnancy Grant	All women in the UK who went to a prenatal appointment received £190; in Scotland, there was no significant change in the odds of low birthweight; there was a significant increase in the odds of scheduling a prenatal care appointment before 25 weeks of gestation (odds ratio 1·10, 95% CI 1·02 to 1·18); the odds of neonatal mortality and emergency caesarean sections increased, maybe due to the 2009 financial crisis and H1N1 influenza pandemic
Reader (2023) ⁵⁷	England and Wales	Quasi-experimental variation in eligibility for Health in Pregnancy Grant, with a regression discontinuity design in the date of birth of the baby	Programme was associated with an increase of 8–12 g in average birthweight and a reduction in low birthweight by 3–6%; preterm births were reduced by 9–11%; for mothers aged ≤21 years, the increase in mean birthweight was 35 g; the primary mechanism of change was likely the cash transfer rather than conditioning on prenatal appointment attendance; findings were more positive than in Leyland et al (2017), ⁴⁵ possibly due to differences in design

APFD=Alaska Permanent Fund Dividend. DID=difference-in-differences. EITC=earned income tax credit. HBPB=Healthy Baby Prenatal Benefit. PROWRA=Personal Responsibility and Work Reconciliation Act. *Preprint papers.

Table 2: Studies on prenatal cash transfers and infant health in high-income countries

welfare benefits of the day and found that the means-tested cash transfers reduced the incidence of infant low birthweight, improved diets through increased consumption of nutritious food, and increased schooling for recipients on low incomes.^{41,65}

Several studies examined the effects of payments from the Alaska Permanent Fund (APF), which was established by statewide referendum in 1976, and through which state residents receive annual payments that vary over time. Based on variation in the timing and value of payments over time, in 2016, Chung and colleagues⁴² found that an added \$1000 in 1982 dollars (\$3720 in 2024) in APF payments increased birthweight by 17·7 g, reduced the incidence of low birthweight by 0·4–0·7 percentage points, and was associated with improved Apgar scores, with effects concentrated among families with lower levels of education. However, a study published in 2024 used more years of variation in APF benefits and found no substantively meaningful effects on birth outcomes, partly because the cash transfers might increase the number of livebirths to parents with greater risk of poor birth outcomes.⁴³ Akee and colleagues^{66–68} observed the effects of per capita cash transfer payments stemming from casino profits among Eastern Cherokee families in North Carolina, USA. Analysing variation in the amount and timing of benefits, they found that among poor households, an additional \$4000 a year was associated with an additional year of schooling by age 21 years, a 22% reduced risk of committing a minor crime for adolescents aged 16 years and 17 years, and substantially improved emotional and behavioural health.^{66–68} Researchers hypothesise that improved parental relationship quality for recipients could be a primary mechanism of change.

The 2021 expanded CTC in the USA

In 2021, the USA briefly transformed its annual CTC into a monthly, near-universal child benefit in line with many of its high-income peers. On the basis of previous Congressional legislation known as the American Family Act, this COVID-19 pandemic era expansion extended the full benefit to families with the lowest incomes, even those with no earnings or income tax liability, and increased the annual value of the credit to \$3000, with an extra \$600 for children younger than 6 years.⁶⁹ The expanded CTC lifted 5·3 million people—including 2·9 million children—out of poverty, slashing child poverty in the USA to a historic low of 5·2%.⁷⁰ While the expanded CTC was in place, the USA shifted from having a child poverty rate that was ranked 30 of 33 OECD countries with available poverty data, to 24 of 33, and only 3 percentage points higher than the median rate.⁷¹ In 2021, food insecurity among households with children fell to the lowest ever recorded (12·5%) going back to 2001,⁷² and financial hardship eased.⁷³ When the monthly benefit expired, these profound positive impacts dissipated, and child poverty more than doubled from 5·2% to 12·4%, the

highest 1 year increase on record.⁷⁴ In 2022, the food insecurity rate among households with children also saw the biggest 1 year increase since 2001, leading to 4·1 million more children in food-insecure households in 2022, compared with 2021.^{72,75}

Given the short duration of the expanded CTC and the fact that it was implemented during the historic public health crisis that was the COVID-19 pandemic, there is little evidence on the health effects of the programme. In a preprint paper, Ruffini¹⁸ analysed the full set of cash transfers made available to American households as Economic Impact Payments and the expanded CTC and found that an additional \$1000 benefit before birth was associated with reduced prevalence of low birthweight of 1·6–3·5% and preterm births by 1·4%. This analysis indicates that cash transfers were linked to increased prenatal care and reduced smoking during the third trimester.¹⁸ These findings are consistent with earlier EITC studies and positive effects were strongest for children conceived before the COVID-19 pandemic.¹⁸ In a contrasting preprint paper, Lyu and colleagues used similar data, but restricted their analysis to parity only among women with the least educational attainment (those with a high school degree or less) and did not assess differential effects of cash transfers for infants conceived before and after the start of the COVID-19 pandemic. The authors found no link between the cash transfers during the COVID-19 pandemic and birthweight.⁴⁴

Several studies examined the mental health of adults with children following the roll-out and then retrenchment of the expanded CTC programme. Cha and colleagues⁷⁶ found that added benefits from the expanded CTC based on age and number of children was linked to reduced psychological distress in adults aged 18–65 years with at least one child, especially in women and Hispanic adults. Another study found that the expanded CTC coincided with a decline in symptoms of anxiety for families reporting annual incomes less than \$35 000.⁷⁷ No significant reductions in depression symptoms were found, although one might not expect such effects from a short-term intervention. Batra and colleagues⁷⁸ found that the expanded CTC was associated with decreased depressive and anxiety symptoms among adults with low incomes and children and with greater anxiety reduction among Black and Hispanic people. In 2023, Kovski and colleagues⁷⁹ analysed a large monthly survey of families receiving the Supplemental Nutrition Assistance Program and found evidence consistent with reduced parental anxiety and depression symptoms linked to the expanded CTC, particularly for non-Hispanic Black parents and those who had received the benefits over a longer duration.⁷⁹ In nationally representative surveys, the relationship between the expanded CTC and mental health improvements was greatest among families on lower incomes, participants aged 18–65 years, women, Black individuals, Hispanic individuals, and those from

other racially and ethnically minoritised groups.^{76–78} Following the expiration of the expanded CTC, there was an increase in the number of adults reporting moderate to severe psychological distress.⁷⁶

Using child cash transfers to improve health

As a national investment, a primary purpose of child cash benefits is to prevent and reduce child poverty and improve family financial security, therefore bolstering family wellbeing and children's health and development. Evidence shows that they are effective in achieving these goals; programmes with the highest coverage rates and higher benefits have the most impact, and universal and near-universal programmes tend to have the highest coverage rates.

The USA briefly joined other high-income nations with a near-universal, unconditional monthly child cash benefit through the expanded CTC. The nation saw large child poverty reductions and concurrent improvements in other indicators of health and wellbeing. These gains were lost when the expanded CTC was not renewed and reverted to a version that did not include refundable tax credits for families on the lowest incomes. Federal lawmakers continue their efforts to reinstate the federal expanded CTC and make it permanent.⁸⁰ In addition, there are now 11 states (including Washington DC)⁸¹ with CTCs that provide a full credit for children in families with no or low earnings. Some programmes are just for young children, others apply to all children, some are just for children in families with the lowest incomes, others apply to the middle class, and one in Massachusetts is universal.⁸¹ In addition, some programmes can deliver the credit in advanced, regular payments, and one in Minnesota mandated in May, 2024, that the state must provide the option of advanced, regular payments.⁸² The largest state CTC to date was signed into law in May, 2024, in Colorado, with a total annual credit of up to \$3200 for children younger than 6 years, and \$2400 for children aged 6–16 years.⁸³

Although meaningful in size and scope, given their restricted reach, state CTCs do not match the impact of the national 2021 expanded CTC. Given its success, the expanded CTC should be restored at the same real value of 2021 (indexed for inflation) and made permanent. In addition, several studies from low-income and high-income countries add to the evidence that benefits delivered during the prenatal period have the potential to improve birth outcomes and thus longitudinal health and development. Given the evidence of the benefits of prenatal cash transfers, policy makers might consider providing cash during pregnancy.

Two of the authors of this Health Policy (HLS and MH) are behind a community-level response to the short-lived success of the expanded CTC. Based on the evidence reviewed in this Health Policy, a universal prenatal and infant cash benefit programme was developed and implemented in January, 2024, in one of the poorest cities

in the USA: Flint, MI. Due to a decrease in income during pregnancy, the additional expenses of a newborn, and the absence of paid maternity leave in the USA, poverty spikes to its highest in the lifecourse at childbirth, beginning prenatally and persisting to at least the first year of life.⁸⁴ In a community-driven effort to eliminate deep infant poverty with a prescription for health, hope, and opportunity, everyone expecting a new baby in Flint receives a one-time unconditional prenatal cash prescription of \$1500 after 20 weeks gestation. Families then receive a \$6000 newborn benefit delivered as \$500 a month after birth until age 1 year. This programme serves all families in a community with a very high child poverty rate. The universal and unconditional design actively conveys the values of dignity, solidarity, deservingness, trust, and love.

In June, 2023, representatives Rosa DeLauro, Suzan DelBene, and Ritchie Torres, with virtually every House Democrat co-sponsoring, reintroduced the American Family Act, a CTC bill⁸⁵ that mirrored the 2021 expanded CTC, but added a new increased credit for infants totalling \$6000.^{86,87} In her campaign proposal to restore the 2021 expanded CTC, Vice President Kamala Harris has included an increased \$6000 fully refundable newborn tax credit, in line with support in the 2023 American Family Act and the post-birth Rx Kids benefit.^{87,88} Vice President Elect JD Vance has also proposed a substantial expansion to the CTC.⁸⁹

Different forms of child cash benefits have been implemented with success in so many countries that it is possible to imagine a broader scale of investment. What if all the world's children were covered by a child cash benefit? Calls for all nations to adopt universal child benefits have been made by family policy experts at the UN^{90,91} and as part of the G20 process,⁹² where it has been argued that such a policy could eradicate extreme child poverty around the globe. Such programmes could follow the approach of Rx Kids by starting with pregnancy and the youngest children, as has also been suggested as part of the G20 consultations,⁹² and add older children over time, as in South Korea and a proposal in Spain.⁴

Conclusion

Child cash benefits independently affect the economic wellbeing and health of families with children and work in concert with the provision of other services, because education and health services underperform in systems without cash benefits. As the USA and other nations consider bold and evidence-based strategies to improve children's health and wellbeing, universal child cash benefits should be at the forefront of the broader economic policy portfolio as an investment in a nation's health, sustainability, and global competitiveness. Universal or near-universal programmes are simple and efficient to implement, are based on decades of global and domestic evidence, and are associated with widespread impact and public support. A healthier world in which all children are covered by a child cash benefit is possible.

Contributors

HLS facilitated the conceptualisation of this Health Policy, took the lead on the investigation and data curation, and took lead in writing the original draft. MH co-facilitated the conceptualisation of this Health Policy, contributed to writing the original draft, and reviewed and editing the manuscript. DH, DR, and ML participated in the conceptualisation of this Health Policy, contributed to the writing of the original draft and reviewed and edited the manuscript.

Declaration of interests

We declare no competing interests.

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